

Coffee Shop Sales Analysis (Excel + Power BI Project)

Project Overview

This project analyzes coffee shop sales data containing **140,000+** records to understand business performance, customer behavior, and product trends. Data cleaning and transformation were performed in Excel, and an interactive dashboard was developed in Power BI to generate actionable insights.

Data Cleaning & Transformation

Extracted Product Size:

- Extracted size (Small, Medium, Large) from product description column

Date Transformation

From the order_date column:

- Extracted Month
- Extracted Day Name (Monday–Sunday)

From the order_time column:

- Extracted Hour

Key Metrics

- Total Revenue: \$698.50K
- Total Orders: 149K
- Total Footfall: 149,116
- Average Order Value: \$4.68
- Average Basket Size: 1.44

Monthly Trend

- Sales vary across months
- Insight: Helps in seasonal planning

Daily Trend

- Certain days generate higher revenue
- Insight: Useful for staff planning

Hourly Trend

- Peak hours identified
- Insight: Optimize operations during peak time

Top Products

- Espresso, Chai Tea, Hot Chocolate
- Insight: High demand → promote these items

Category Contribution

- Coffee dominates revenue
- Insight: Core business driver

Size-wise Sales

- Large size generates more revenue
- Insight: Opportunity for premium pricing

Location Analysis

- High-performing locations identified
- Insight: Focus marketing efforts on top locations

Key Business Insights

- Coffee category drives most revenue
- Peak hours show strong morning demand
- Large size products are more profitable
- Some products underperform
- Low basket size → opportunity for combo deals

The analysis shows that the coffee category is the main driver of revenue, meaning most customers visit primarily for coffee products. To take advantage of this, the business can introduce premium options and loyalty programs to increase repeat purchases. The data also highlights strong morning demand, so increasing staff during peak hours and offering morning deals can improve service and boost sales. Large-size products generate higher revenue, indicating customers prefer better value, so promoting size upgrades and combo offers can increase average order value. Some products are underperforming, suggesting they should be improved, discounted, or replaced with more popular items. The low basket size shows customers usually buy only one or two items, so introducing combo deals and add-on suggestions can increase sales per order. Focusing on high-performing products and optimizing the menu can also improve efficiency and reduce waste. Overall, these strategies can help increase revenue, improve customer experience, and drive business growth.



Conclusion

This project demonstrates strong skills in data cleaning, transformation, and visualization using Excel and Power BI. The analysis provides meaningful insights that can help improve business decisions, increase revenue, and optimize operations.